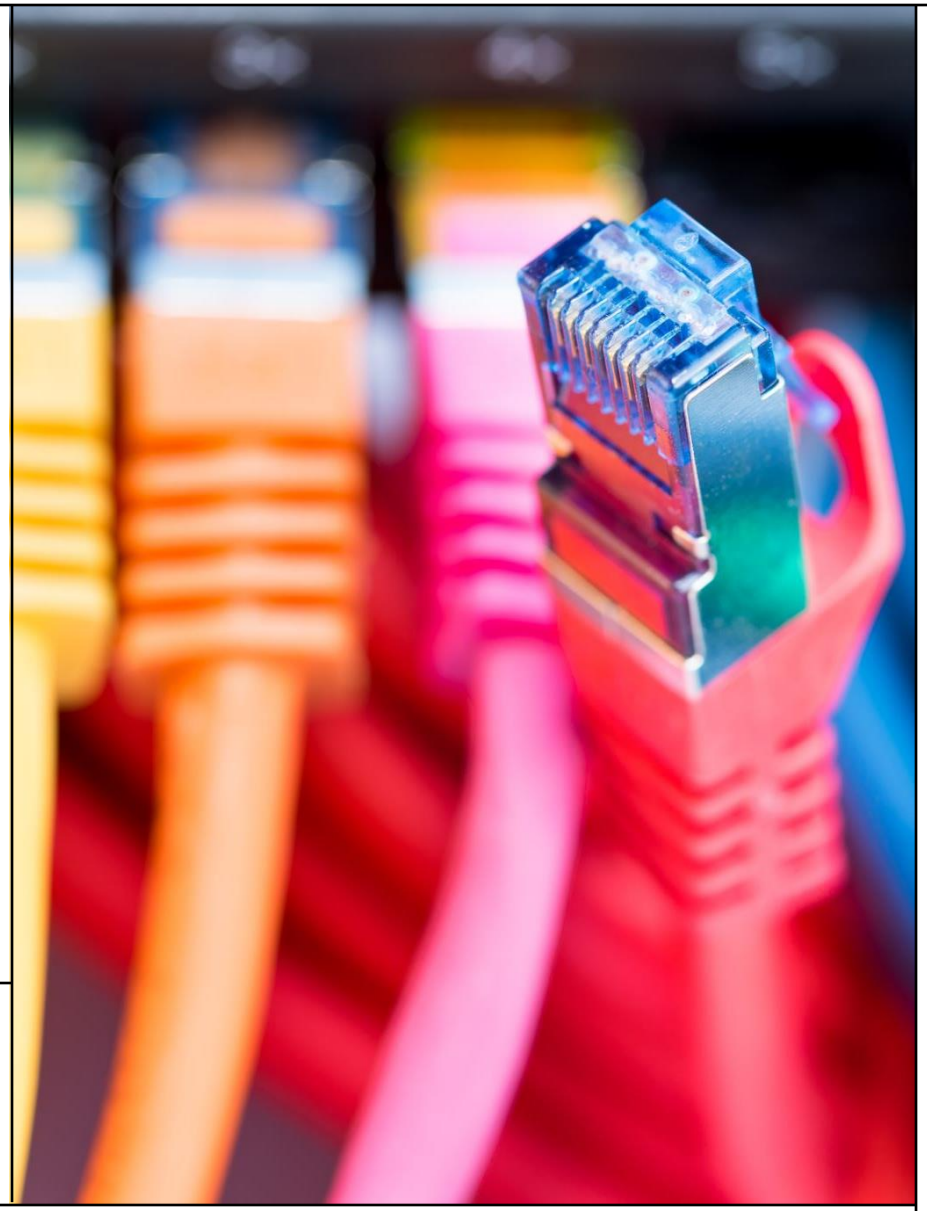
VM network connection types

In this presentation we will review bridged, NAT, Host-Only, Custom VM network connection types and their Pros and cons



Bridged network connection type

- Connect the VM (Virtual Machine) to the physical network
- The VM has it’s own IP Address

- PROS :

  -> Handles direct access between the VM and the Network

  -> Allows testing environments and routes

  -> Allows Snapshots (Ability to revert the state of the VM)

- CONS

  -> Needs a proper network configuration to avoid IP conflicts

  -> Exposed to attacks due to its exposure to the network

  -> Often requires a manually assigned IP address to avoid conflicts

NAT (Network Address Translation) Network connection type

- Connect the VM (Virtual Machine) to the Internet trough the host IP address
- NAT act as a private network
- PROS :

  -> Allows the VM to use the Host IP address to connect to the internet

  -> Allows Internet access without exposing the VM to attackers

  -> Ease of setup and maintenance
- CONS

  -> Make the VM not accessible from the internet

  -> Unable to test connectivity with external servers

Host Only Network connection type

- Connect the VM (Virtual Machine) to the Host trough a custom private network
- Act as a private network completely closed to the outside
- PROS :

  -> Provides a high level of security

  -> Allows testing without Internet
- CONS

  -> Make the VM not accessible from the internet

  -> Limited use cases for real life production setup

Custom Network connection type

- A custom network means that it is made for a specific use case
- It can combine Bridged / NAT / Host Only Network connection types
- Act as a private network completely closed to the outside
- PROS :

  -> Provides a high level of security

  -> Made for specific and advanced networking scenario

  -> Unlimited use cases for real life production setup
- CONS

  -> Hard to setup and to maintain

  -> Often comes at a high price

Chose Your network connection type

Depending on your specific use case chose carefully putting in correlation the risk level and the desired utility

Type	Access	Risk	Utility
Bridged	Straight to physical Network	Low	Allows effitient testing
NAT	Trough the Host's IP address	Moderated	Provides internet access without exposure
Host Only	Trough the host and VM	High	Secured and isolated testing
Custom	Depends on the need	Depends	Advanced Network setups settings and testing

Conclusion

When we look at a new project, we can easily identify needs for the network setup

-> How much risk you can allow (how much can lose vs the cost of the setup)

-> Can your issue be easily solved with an existing configuration

-> How much your system needs to be reliable

For most projects, A NAT network might be enough as it provides a great balance for security while keeping access to the internet.

Thank you for your attention

I hope you liked today’s presentation

References :

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